

Jacob Keller

jkeller@olin.edu | (949) 560-7253 | [linkedin.com/in/jacob-keller-profile](https://www.linkedin.com/in/jacob-keller-profile) | jacobkeller.me

Education

Olin College of Engineering – Needham, MA

May 2028

- BS, Electrical and Computer Engineering
- GPA: 4.0
- Relevant Coursework: Software Systems (**C**), Software Design (**Python**), Electronics, Modeling and Simulation (**MATLAB**), Principles of Integrated Engineering, Linear Algebra, Multivariable Calculus, Differential Equations

Experience

EV Engineering Intern, Formosa EV – Orange, CA

Mar 2025 – Aug 2025

- First intern and early employee at startup developing modern EV conversion components
- Designed high-voltage bus bars and sheet metal brackets in **Onshape** for motor, battery, and radiator integration.
- Flashed and tested firmware updates on MCU/SoC processors, documenting errors to support development team
- Adapted classic car systems (e.g., throttle linkage) to modern electronic controls, reducing cost with **3D-printed** parts

Low Voltage Lead, Olin Formula SAE Electric – Needham, MA

May 2025 – Present

- Led 5 engineers across 9 projects spanning PCB design (**KiCAD**), **C** firmware, and **Python** telemetry (in progress)
- Created technical project descriptions and a detailed timeline to ensure on-time completion of team deliverables

Integration Engineer, Olin Formula SAE Electric – Needham, MA

Sept 2024 – May 2025

- Developed embedded firmware in **C** for ATmega16M1 microcontroller: implemented CAN bus messaging, ADC readings, SPI-driven display and IO expander, and cooling control (PWM). Controls team's power distribution unit
- Created **Python** wireless telemetry system: serialized CAN messages, transmitted via UART radio, then deserialized and displayed in existing CAN dashboard, enabling drive-time diagnostics for the first time in team history

Information Technology Helpdesk Staff, Olin College – Needham, MA

Sept 2024 – Present

- Delivered technical support to a diverse user base of 450+ students, faculty, and staff as a Dell Certified Technician
- Created clear communication protocols for complex troubleshooting procedures
- Diagnosed and resolved hardware/software issues across multiple operating systems and laptop configurations

Projects

Pedestrian Detection Algorithm, jacobkeller.me/ped

Dec 2024

- Built a **MATLAB** algorithm to detect the presence of a person in an image using principal component analysis (PCA)
- Achieved 78% classification accuracy, demonstrating PCA's effectiveness in improving computational efficiency while maintaining reasonable accuracy for pedestrian detection
- Presented findings to Olin College faculty and students through a professional academic poster

Carbon Dioxide Measurement Station, jacobkeller.me/co2

Aug 2024

- Designed a low-cost solar-powered device for measuring carbon cycling in scientific research
- Prototyped and refined electronic design through breadboard testing and final soldered assembly
- Developed **Python** software for an ESP32 microcontroller using PWM, SPI, I2C, and UART protocols to interface with environmental sensors, a real-time clock, and a linear actuator, while storing collected data on a microSD card

Skills

Programming: Python, C, C++, MATLAB

Software: Git, Linux, OnShape, Solidworks, KiCAD

Manufacturing: 3D Printing, Welding (TIG, Handheld Laser)